6 Second Order Circuits

INTRODUCTION

- We have discussed circuits containing single storage elements, such as a capacitor or an inductor, in chapter 5. Such circuits are termed as first-order because the order of differential equations are first-order.
- In this chapter, we will learn about circuits containing two energy storing elements (L and C) and responses of these circuits are described by second-order differential equations, so the circuit is a second-order circuit.
- The examples of second-order circuits are series RLC circuits and parallel RLC circuits and these circuits consist of three kinds of passive elements like resistor, capacitor and inductor.
- A second-order circuit may be an operational amplifier (Op-amp) circuit with two energy storage elements.
- A second-order circuit, like a first-order circuit, can have several dependent and independent sources and resistors.
- In this chapter, we will learn about source free series RLC circuit, source free parallel RLC circuit, with source series RLC circuit and with source parallel RLC circuit.

Grey Matter Alert!

A second-order circuit consists of resistors and the equivalent of two energy storing elements (L and C).

SOURCE FREE SERIES RLC CIRCUIT

- Series RLC circuit is excited by the initial energy stored in the capacitor and inductor.
- The energy is represented by the initial inductor current (I_0) and the initial voltage capacitor (v_c) as shown in the figure below.

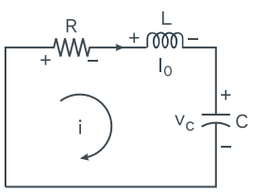


Fig. 6.1 Source Free Series RLC Circuit

Applying KVL in the loop of the circuit, we get,

$$-iR - L\frac{di}{dt} - v_C = 0$$

$$iR + L\frac{di}{dt} + v_C = 0$$

Grey Matter Alert!

The ratio of damping factor (or Neper frequency) to the resonant frequency is known as damping ratio(ξ).

$$\xi = \frac{\alpha}{\omega_0}$$

But, $i = C\frac{dv_C}{dt}$

$$LC\frac{d^2v_C}{dt^2} + RC\frac{dv_C}{dt} + v_C = 0$$

$$\frac{d^2v_C}{dt^2} + \frac{R}{L}\frac{dv_C}{dt} + \frac{v_C}{LC} = 0$$

By method of the differential operator,

putting $\frac{d^2}{dt^2} = D^2$, $\frac{d}{dt} = D$

$$\left[D^2 + \frac{R}{L}D + \frac{1}{LC}\right]v_c = 0$$

Therefore, the characteristic equation is:

$$D^2 + \frac{R}{L}D + \frac{1}{LC} = 0$$

$$\text{roots} = \frac{-\frac{R}{L} \pm \sqrt{\frac{R^2}{L^2} - \frac{4}{LC}}}{2}$$

$$= -\frac{R}{2L} \pm \sqrt{\left(\frac{R}{2L}\right)^2 - \frac{1}{LC}}$$

$$= -\alpha \pm \sqrt{\alpha^2 - \omega_0^2}$$

Where α = damping factor = $\frac{R}{2L}$

ω_0 = natural frequency = $\frac{1}{\sqrt{LC}}$

- In a second-order circuit, we generally observe oscillations in the output.
- The damping factor is related to resistance and hence damps out these oscillations.
- Natural frequency is the frequency of oscillation if there is no damping provided. In such a case, oscillations can go on for infinite time without requiring any energy from an external source. Such oscillations are called self-sustained oscillations.

Rack Your Brain

What is the transient analysis of the RLC circuit?

Definitions

Second-order circuits:
"A second-order circuit is characterized by a second-order differential equation, and it consists of resistors and two equivalent energy storing elements, capacitors and inductors".

Case-1:

If $\alpha > \omega_0$,

Roots $= -\alpha \pm \sqrt{\alpha^2 - \omega_0^2}$

Roots are real and distinct, response is overdamped roots = s_1, s_2

$$v_C = C_1 e^{s_1 t} + C_2 e^{s_2 t};$$

C_1 and C_2 are arbitrary constants and can be derived by using initial conditions.

Grey Matter Alert!

The voltage across the capacitor is always continuous i.e. $v(0^+) = v(0^-)$

Case-2:

If $\alpha = \omega_0$,

Roots $= -a$

Roots are real and equal, response is critically damped.

$$v_C = (C_1 + C_2 t)\, e^{-at}$$

Case-3:

If $\alpha < \omega_0$,

Roots $= -\alpha \pm \sqrt{\alpha^2 - \omega_0^2}$

$$= -\alpha \pm j\sqrt{\omega_0^2 - \alpha^2}$$

$$= -\alpha \pm j\omega_d \,(\text{complex})$$

Roots are complex conjugate, and the response is underdamped.

α : damping factor

ω_d: damping frequency

$$v_C(t) = e^{-\alpha t}\left[C_1 \cos \omega_d t + C_2 \sin \omega_d t\right]$$

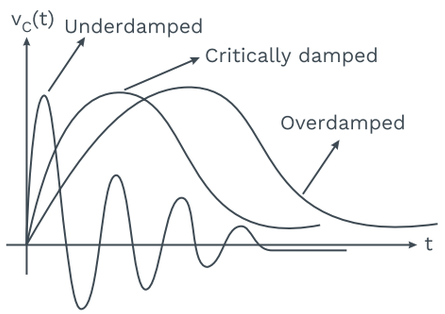


Fig. 6.2 Response of Three Degrees Damping

- $\cos\omega_d t$ and $\sin\omega_d t$ are oscillatory terms. So, the nature of the response is oscillatory.
- e^{-at} : Reduces amplitude of oscillations with time.

Frequency = $\omega_d = \sqrt{\omega_0^2 - \alpha^2}$

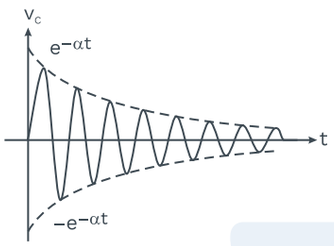


Fig. 6.3 Underdamped Response

- The overdamped response is slower than the critically damped response.
- As damping increases, the speed of response reduces.
  If R = 0, α = 0 no damping (undamped system), the amplitude of oscillations remains constant and never reduces.

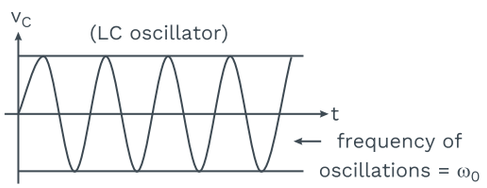


Fig. 6.4 Undamped Response

- For any parameter in the circuit, the equation of the response remains the same based on the nature of damping. But the value of arbitrary constants change.

SOURCE FREE PARALLEL RLC CIRCUIT

- A parallel RLC circuit is excited by the initial energy stored in the capacitor and inductor.
- The energy is represented by the initial capacitor voltage v_c and initial inductor current I_0 as shown in the figure below.

Rack Your Brain

What is the transient analysis of the RLC circuit?

- Parallel RLC circuits are used in practical applications, for example, in communications networks and filter designs.

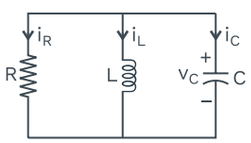


Fig. 6.5 A Source-Free Parallel RLC Circuit

Applying KCL in the circuit,

$$i_R + i_L + i_C = 0$$

$$\frac{v_C}{R} + i_L + C\frac{dv_C}{dt} = 0$$

Differentiating with respect to t, we get,

$$\frac{1}{R}\frac{dv_C}{dt} + \frac{di_L}{dt} + C\frac{d^2v_C}{dt^2} = 0$$

$$\frac{di_L}{dt} = \frac{v_C}{L} \qquad [\because v_C = v_L]$$

$$\frac{1}{R}\frac{dv_C}{dt} + \frac{v_C}{L} + C\frac{d^2v_C}{dt^2} = 0$$

$$\frac{d^2v_C}{dt^2} + \frac{1}{RC}\frac{dv_C}{dt} + \frac{v_C}{LC} = 0$$

By method of the differential operator:

$$\left[D^2 + \frac{1}{RC}D + \frac{1}{LC}\right]v_C = 0$$

$$\text{Roots} = -\frac{1}{2RC} \pm \sqrt{\left(\frac{1}{2RC}\right)^2 - \frac{1}{LC}}$$

$$\text{Damping factor} \Rightarrow \alpha = \frac{1}{2RC}$$

$$\text{Natural frequency} = \frac{1}{\sqrt{LC}}$$

Case-1:

If $\alpha > \omega_0$ (overdamped) roots are real and distinct.

Roots = s_1, s_2

$$v_C(t) = C_1e^{s_1t} + C_2e^{s_2t}$$

Note: C_1 and C_2 are derived using initial conditions.

Rack Your Brain

What is the significance of undamped natural frequency and damped natural frequency?

Case-2:

If $\alpha = \omega_0$ (critically damped), roots are real and equal.

Roots = $-a$

$$v_C(t) = (C_1 + C_2 t)e^{-at}$$

Case-3: If $\alpha < \omega_0$,

(underdamped) roots are complex conjugate $\omega_d = \sqrt{\omega_0^2 - \alpha^2}$

$$v_C(t) = e^{-\alpha t}(C_1 \cos \omega_d t + C_2 \sin \omega_d t)$$

Everything remains the same except the value of α.

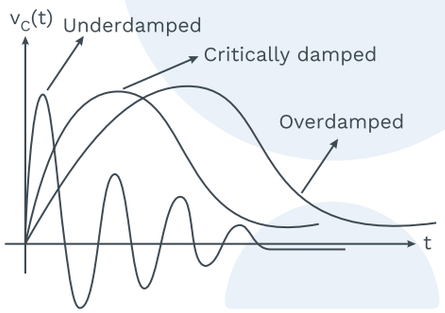


Fig. 6.6 Response of Three Degrees Damping

Note:

In series RLC, $\alpha \propto R$

In parallel RLC, $\alpha \propto G = \frac{1}{R}$

Summary of steps:

- Find initial condition $i_L(0^-)$ and $v_C(0^-)$
- Draw circuit for $t > 0$
- Find α and ω_0.
- Judge the nature of damping and write the equation for the desired parameter.
- Find the arbitrary constant using current and voltage across the element given.

Grey Matter Alert!

- When the roots of the characteristic equation of the circuit is unequal and real, then the response is overdamped.
- The critical damped response is obtained when the roots of characteristics equations are real and equal.
- When the roots of the characteristics equation of the circuit is complex, then the response is underdamped.

TIME CONSTANT

For an underdamped and critically damped system.

$$\tau_1 = \tau_2 = \frac{1}{\alpha}$$

For an overdamped system, we need to determine the roots of the characteristic equation (s_1, s_2).

$$\tau_1 = \frac{1}{|s_1|}, \quad \tau_2 = \frac{1}{|s_2|}$$

Example 1:

For the circuit shown below, determine i_L and v_C for $t \geq 0$.

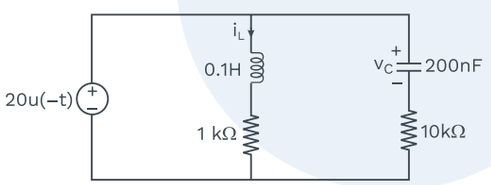


Fig. 6.7

Sol:

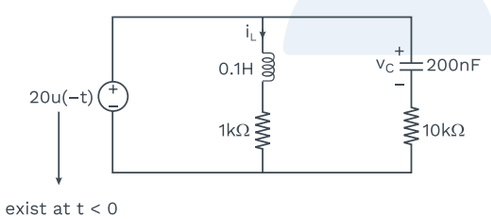


Fig. 6.8

At $t = 0^-$, circuit in steady-state
L : short circuit, C : open circuit

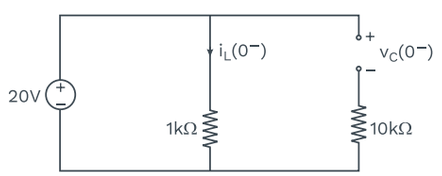


Fig. 6.9

$$i_L(0^-) = \frac{20}{1K} = 20 \text{ mA}$$

$$v_C(0^-) = 20 \text{ volts}$$

Rack Your Brain

What is the significance of the time constant of the parallel RLC circuit?

For t > 0, source = 0 (short circuit)

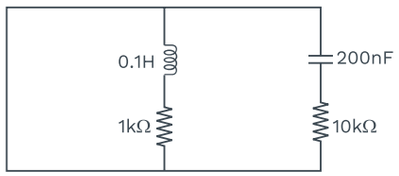


Fig. 6.10

Since, both branches (RC and RL) are shorted, the response of both can be determined separately. If the source branch were open, the same current would circulate throughout the circuit.

$$i_L(t) = 20e^{-t/\tau_1}\text{mA}$$

$$\tau_1 = \frac{0.1}{1K} = 0.1\ \text{m sec}$$

$$i_L(t) = 20e^{-10000t}\text{mA}$$

$$v_C(t) = 20e^{-t/\tau_2}\text{V}$$

$$\tau_2 = 10\,\text{k}\Omega \times 200\,\text{nF} = 2\,\text{m sec}$$

$$v_C(t) = 20e^{-500t}\text{V}$$

Example 2:

The circuit shown below has reached a steady-state at $t = 0^-$. If the switch move from position a to b at t = 0, determine the current i(t) for t > 0.

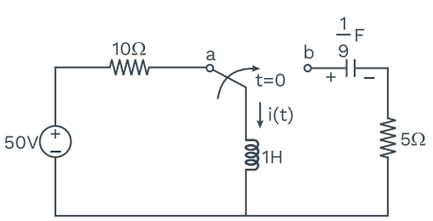


Fig. 6.11

Sol:

At $t = 0^-$, the switch is closed, and the circuit is in the steady-state.
L: short circuit, C: open circuit

$$i_L(0^-) = \frac{50}{10} = 5\text{A}$$

$v_C(0^-) = 0\text{V}$ (no source)

For t > 0;

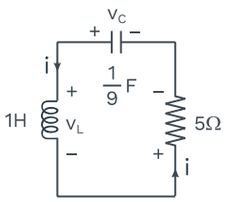


Fig. 6.12

For series RLC circuit,

$$\alpha = \frac{R}{2L} = \frac{5}{2 \times 1} = 2.5$$

$$\omega_0 = \frac{1}{\sqrt{LC}} = 3$$

$\alpha < w_0$: underdamped

$i(t) = e^{-at} (C_1 \cos \omega_d t + C_2 \sin \omega_d t)$

$$\omega_d = \sqrt{\omega_0^2 - \alpha^2} = \sqrt{2.75} = 1.658$$

At $t = 0^+$, $i(t) = 5A$

$e^{-a \times 0} (C_1 \cos 0 + C_2 \sin 0) = 5$

$C_1 = 5$

By KVL,

$v_L - v_C + iR = 0$

At $t = 0^+$,

$v_L - 0 + 5 \times 5 = 0$

$v_L = -25V$

$$v_L = L\frac{di}{dt} = -\alpha e^{-\alpha t}\left(C_1 \cos \omega_d t + C_2 \sin \omega_d t\right)$$

$$+e^{-\alpha t}\left[-\omega_d C_1 \sin \omega_d t + C_2 \omega_d \cos \omega_d t\right]$$

At $t = 0^+$,

$-25 = -\alpha (C_1) + 1[C_2 \omega_d]$

$-25 = -2.5 \times 5 + C_2 \omega_d$

$$C_2 = -\frac{12.5}{\omega_d} = -\frac{12.5}{1.658} = -7.54$$

$$i(t) = e^{-2.5t}\left[5 \cos \omega_d t - 7.54 \sin \omega_d t\right] A$$

Rack Your Brain

What is the complete response of an RLC circuit?

Example 3:

Find v(t) for t > 0 in the RLC circuit shown below.

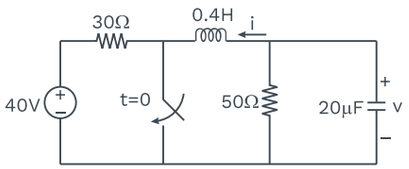


Fig. 6.13

Sol:

At t = 0⁻, the switch is open.

The circuit is in the steady-state.

L: short circuit, C: open circuit

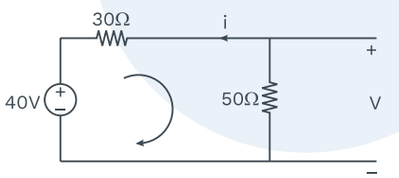


Fig. 6.14

$$i(0^-) = -\frac{40}{80} = -0.5A$$

$$v(0^-) = 40 \times \frac{50}{80} = 25V$$

For t > 0,

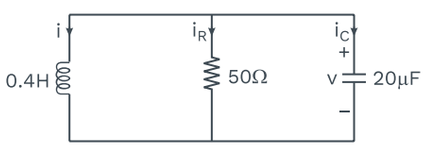


Fig. 6.15

$$\alpha = \frac{1}{2RC} = \frac{1}{2 \times 50 \times 20 \times 10^{-6}} = 500$$

$$\omega_0 = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{0.4 \times 20 \times 10^{-6}}} = \frac{1000}{2\sqrt{2}} = \frac{500}{\sqrt{2}}$$

$$\text{Roots} = -\alpha \pm \sqrt{\alpha^2 - \omega_0^2}$$

$$= -500 \pm \sqrt{500^2 - 500^2 / 2}$$

$$= -500 \pm \frac{500}{\sqrt{2}}$$

$$= -853.55, -146.45$$

$s_1 = -853.55$

$s_2 = -146.45$

$$v(t) = C_1 e^{s_1 t} + C_2 e^{s_2 t}$$

At $t = 0^+$

$v(t) = 25V$

$25 = C_1 + C_2$ ...(i)

For second equation use capacitor current,

$$i_C(t) = C\frac{dv(t)}{dt} = C\left[C_1 s_1 e^{s_1 t} + C_2 s_2 e^{s_2 t}\right]$$

By KCL,

$i_C + i_R + i = 0$

At $t = 0^+$

$$i_C(0^+) + \frac{V_C(0^+)}{R} + i(0^+) = 0$$

$$i_C(0^+) + \frac{25}{50} + (-0.5) = 0$$

$i_C(0^+) = 0$ A

$$C\left[C_1 s_1 e^{s_1 x 0} + C_2 s_2 e^{s_2 x 0}\right] = 0$$

$$C_1 s_1 + C_2 s_2 = 0$$

$C_1(-853.55) + C_2(-146.45) = 0$

$C_2 = -5.828\ C_1$ ...(ii)

From equations i) and ii), we get,

$C_1 = -5.177$

$C_2 = 30.177$

$$v(t) = -5.177e^{-853.55t} + 30.177e^{-146.45t}V$$

Grey Matter Alert!

The inductor current is always continuous. i.e. $i(0^+) = i(0^-)$

WITH SOURCE SERIES RLC CIRCUIT (STEP RESPONSE OF SECOND-ORDER CIRCUITS)

- The step response with source, series RLC circuit is obtained by the abrupt application of a dc source. Here we want to find voltage (v) due to the sudden application of a dc voltage (V_s).

- Consider the series RLC circuit as shown below.

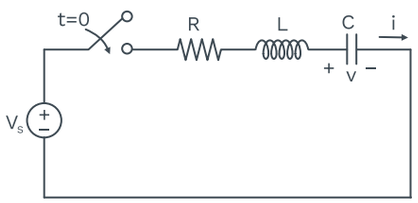


Fig. 6.16 Step Voltage Applied to Series RLC Circuit

- Two initial conditions must be known to evaluate the two arbitrary constants.
  Under steady-state:
  C : open circuit
  L : short circuit

$$v(\infty) = V_S \qquad i_L(\infty) = 0$$
$$v_R(\infty) = 0 \qquad v_L(\infty) = 0$$

Applying KVL around the loop for t > 0.

$$-V_s + iR + L\frac{di}{dt} + v = 0$$

But, $i = C\frac{dv}{dt}$

$$-V_s + RC\frac{dv}{dt} + LC\frac{d^2v}{dt^2} + v = 0$$

$$\frac{d^2v}{dt^2} + \frac{R}{L}\frac{dv}{dt} + \frac{v}{LC} = \frac{V_s}{LC} \qquad \text{...(i)}$$

The solution of equation i) has two components;

i) the transient response $v_t(t)$

ii) the steady state response $v_{ss}(t)$

$$v(t) = v_t(t) + v_{ss}(t) \qquad \text{...(ii)}$$

Case-1: $\alpha < \omega_0$ (underdamped)

$$v(t) = v(\infty) + e^{-\alpha t}[C_1\cos\omega_d t + C_2\sin\omega_d t] \qquad \text{...(iii)}$$

Compute C_1, C_2 using initial conditions.

Case-2: $\alpha = \omega_0$ (critically damped)

$$v(t) = v(\infty) + [C_1 + C_2 t]e^{-\alpha t} \qquad \text{...(iv)}$$

Case-3: $\alpha > \omega_0$ (overdamped)

$$s_1 = -\alpha + \sqrt{\alpha^2 - \omega_0^2}$$

$$s_2 = -\alpha - \sqrt{\alpha^2 - \omega_0^2}$$

$$v(t) = v(\infty) + [C_1 e^{s_1 t} + C_2 e^{s_2 t}] \quad \ldots(v)$$

When computing C_1 and C_2 using initial conditions, make sure $v(\infty)$ is included in the equation:

$$\alpha = \frac{R}{2L}$$

$$\omega_0 = \frac{1}{\sqrt{LC}}$$

$$\omega_d = \sqrt{\omega_0^2 - \alpha^2}$$

$$s_{1,2} = -\alpha \pm \sqrt{\alpha^2 - \omega_0^2}$$

Grey Matter Alert!

Two circuits are called to be duals of one another, if the characteristics equations of both the circuits are the same with dual quantities interchanged.

Response:

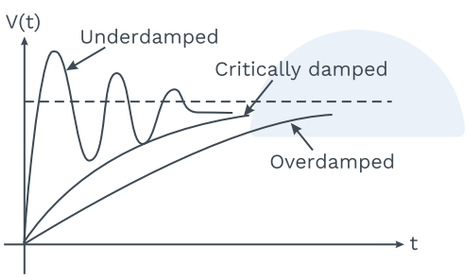


Fig. 6.17 Response of Three Degrees Damping

- Overdamped has the slowest response.
- All responses settle down at the steady-state value.

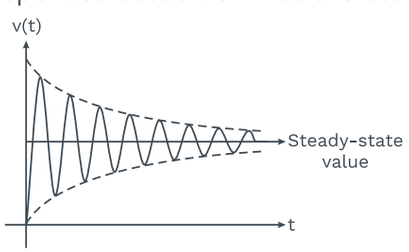


Fig. 6.18 Underdamped Response

Rack Your Brain

What is the significance of the time constant of the series RLC circuit?

WITH SOURCE PARALLEL RLC CIRCUIT

- The step response with source parallel RLC circuit is obtained by the sudden application of a dc source. Here we want to find i_L due to the sudden application of a dc.
- Consider the parallel RLC circuit as shown below.

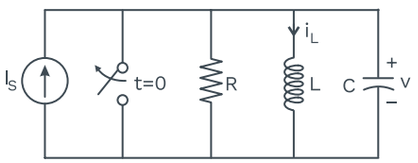


Fig. 6.19 Parallel RLC Circuit with an Applied Current (I_s)

At t = 0, the switch is open, so the current from the source starts flowing into R, L, C network.
Under steady-state

L: short circuit, C: open circuit

Entire current flow through L.

$$i_L(\infty) = I_s, \quad v(\infty) = 0$$

$$v_R(\infty) = 0, \quad i_R(\infty) = 0$$

Applying KCL at the top node of the above circuit for t > 0;

$$\frac{v}{R} + i + C\frac{dv}{dt} = I_s$$

But, $v = L\frac{di}{dt}$

$$\frac{L}{R}\frac{di}{dt} + i + LC\frac{d^2i}{dt^2} = I_s$$

$$\text{Or } \frac{d^2i}{dt^2} + \frac{1}{RC}\frac{di}{dt} + \frac{i}{LC} = \frac{I_s}{LC} \qquad \text{...(i)}$$

The solution of equation i) is consisting of two components:

i) the transient response $i_t(t)$

ii) the steady state response $i_{ss}(t)$

$$i(t) = i_t(t) + i_{ss}(t) \qquad \text{...(ii)}$$

Case-1: $\alpha < \omega_0$ (under damped)

$$i(t) = i(\infty) + e^{-\alpha t}[C_1\cos\omega_d t + C_2\sin\omega_d t] \qquad \text{...(iii)}$$

Compute C_1, C_2 using initial conditions.

Case-2: $\alpha = \omega_0$ (critically damped)

$$i(t) = i(\infty) + [C_1 + C_2 t]e^{-\alpha t} \quad \text{...(iv)}$$

Case-3: $\alpha > \omega_0$ (over damped)

$$s_1 = -\alpha + \sqrt{\alpha^2 - \omega_0^2}$$

$$s_2 = -\alpha - \sqrt{\alpha^2 - \omega_0^2}$$

$$i(t) = i(\infty) + [C_1 e^{s_1 t} + C_2 e^{s_2 t}] \quad \text{...(v)}$$

When computing C_1 and C_2 using initial conditions, make sure $I(\infty)$ is included in the equation

$$\alpha = \frac{1}{2RC}, \ \omega_0 = \frac{1}{\sqrt{LC}}$$

$$\omega_d = \sqrt{\omega_0^2 - \alpha^2}$$

$$s_{1,2} = -\alpha \pm \sqrt{\alpha^2 - \omega_0^2}$$

Rack Your Brain

What is the difference between RLC and LCR circuits?

Response:

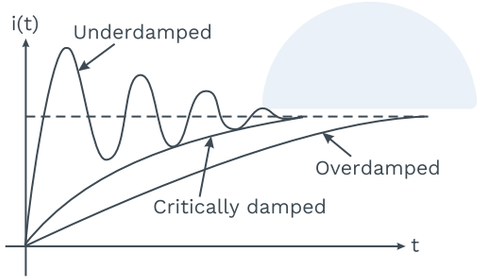


Fig. 6.20: Response of Three Degrees Damping

Example 4:

For the circuit shown below, determine voltage v(t) across the capacitor and current i(t) through the inductor, for t > 0, if the value of R = 5Ω.

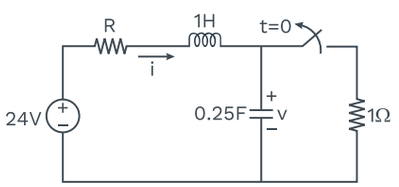


Fig. 6.21

Sol:

At t = 0⁻, switch: closed, the circuit is in the steady-state.

C: open circuit, L: short circuit

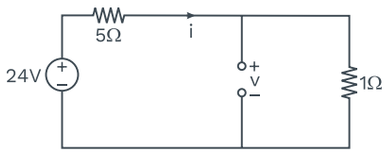


Fig. 6.22

$$i(0^-) = \frac{24}{5+1} = 4\ A$$

$$V(0^-) = 24 \times \frac{1}{5+1} = 4\ V$$

For steady-state ($t \to \infty$)

Switch: open, L: short circuit, C: open circuit

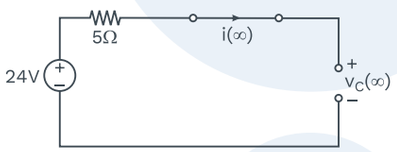


Fig. 6.23

$i(\infty) = 0A$

$v_C(\infty) = 24$ volts

Series circuit, $i_L = i_C$

$$\alpha = \frac{R}{2L} = \frac{5}{2 \times 1} = 2.5$$

$$\omega_0 = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{1 \times 0.25}} = 2\,rad/sec$$

$\alpha > \omega_0$: overdamped

Roots $= -\alpha \pm \sqrt{\alpha^2 - \omega_0^2}$

$= -2.5 \pm 1.5$

$= -4, -1$

$$v(t) = v(\infty) + C_1 e^{s_1 t} + C_2 e^{s_2 t}$$

$$= 24 + C_1 e^{-4t} + C_2 e^{-t}$$

At $t = 0^+$,

$v(t) = 4$ V

$24 + C_1 + C_2 = 4$

$C_1 + C_2 = -20$ ...(i)

$$i(t) = C\frac{dv(t)}{dt} = C \times \left(0 + C_1 s_1 e^{s_1 t} + C_2 s_2 e^{s_2 t}\right)$$
$$= C \times \left(-4C_1 e^{-4t} - C_2 e^{-t}\right)$$

At $t = 0^+$

$i(0^+) = 4$ A

$4 = 0.25 \times (-4C_1 - C_2)$

$4C_1 + C_2 = -16$ ...(ii)

From equations i) and ii), we get,

$3C_1 = 4$

$$C_1 = \frac{4}{3}$$

$$C_2 = -\frac{64}{3}$$

$$v(t) = 24 + \frac{4}{3}e^{-4t} - \frac{64}{3}e^{-t}\,V$$

$$i(t) = \left(-\frac{16}{3}e^{-4t} + \frac{64}{3}e^{-t}\right) \times 0.25$$

$$= -\frac{4}{3}e^{-4t} + \frac{16}{3}e^{-t}\ A$$

Rack Your Brain

What is the Q factor in the RLC circuit?

Example 5:

For the network shown below. Determine the current i(t) for t > 0.

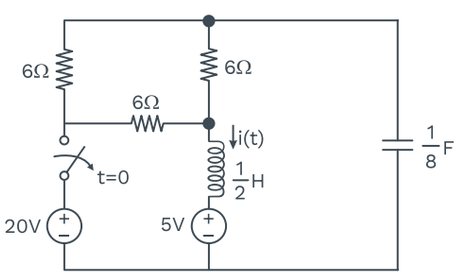


Fig. 6.24

Sol:

At $t = 0^-$, circuit in the steady-state.

L = short-circuit

C = open-circuit

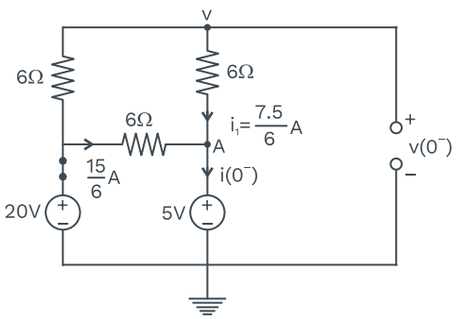


Fig. 6.25

Applying KCL at node v, we get,

$$\frac{v-20}{6}+\frac{v-5}{6}=0$$

$$\frac{v}{3}=\frac{20}{6}+\frac{5}{6}$$

$$\frac{v}{3}=\frac{25}{6}$$

$$v = 12.5V$$

$$\therefore \quad v(0^-) = 12.5V$$

$$i_1=\frac{12.5-5}{6}=\frac{7.5}{6}A$$

Applying KCL at node A, we get,

$$i(0^-)=\frac{15}{6}+\frac{7.5}{6}$$

$$i(0^-)=\frac{22.5}{6}A$$

For t > 0, the switch is open.
The circuit can be redrawn as shown below,

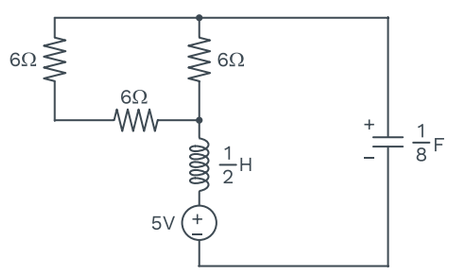


Fig. 6.26

R_{eq} = (6 + 6) || 6 = 4Ω

$$i(0^-) = i(0^+) = \frac{22.5}{6}\,A$$

$$v_c(0^-) = v_c(0^+) = 12.5V$$

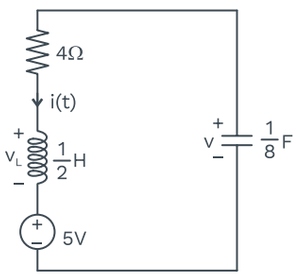


Fig. 6.27

Under steady-state:

C : open circuit, L = short circuit

$i(\infty) = 0A,\ v(\infty) = 5V$

$$\alpha = \frac{R}{2L} = \frac{4}{2 \times \frac{1}{2}} = 4$$

$$\omega = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{\frac{1}{2} \times \frac{1}{8}}} = 4\,rad/sec$$

$\alpha = \omega_0$(critically damped)

$i(t) = i(\infty) + [C_1 + C_2 t]e^{-\alpha t}$

$i(t) = 0 + (C_1 + C_2 t)e^{-4t}$

$$i(t) = (C_1 + C_2 t)e^{-4t} \qquad ...(i)$$

At $t = 0^+$

$i(0^+) = (C_1 + 0) = C_1$

$$C_1 = \frac{22.5}{6} = 3.75$$

Applying KVL in above circuits, we get,

$-5 - v_L - i(t)R + v = 0$

At $t = 0^+$

$$-5 - v_L(0^+) - \frac{22.5}{6} \times 4 + 12.5 = 0$$

$$-v_L(0^+) = 5 + \frac{22.5 \times 4}{6} - 12.5$$

$$v_L(0^+) = -7.5V$$

Rack Your Brain

What is the bandwidth of the RLC circuit?

$$v_L = L\frac{di}{dt}$$

$$v_L = L\frac{d}{dt}(C_1 + C_2 t)e^{-4t}$$

$$v_L = L\left[-4C_1 e^{-4t} + C_2\left(-4t\,e^{-4t} + e^{-4t}\right)\right]$$

$$v_L = L\left[C_2\, e^{-4t} + (-4e^{-4t})(C_1 + C_2 t)\right]$$

$$v_L = \frac{1}{2}\left[C_2 e^{-4t} - 4e^{-4t}(C_1 + C_2 t)\right]$$

At $t = 0^+$

$$-7.5 = \frac{1}{2}[C_2 - 4C_1]$$

$$C_2 - 4C_1 = -15$$

$$C_2 = -15 + 4C_1$$

$$= -15 + 4 \times \frac{22.5}{6} = 0$$

Putting the value of C_1 and C_2 in equation i), we get:

$i(t) = 3.75\ e^{-4t}A$

Example 6:

For the circuit shown in the figure, determine i_L and v_L for $t \geq 0$.

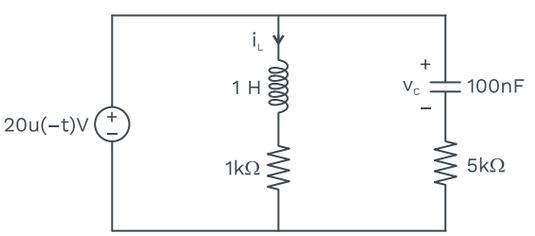


Fig. 6.28

Sol:

$$u(-t) = \begin{bmatrix} 1\ , & t < 0 \\ 0\ , & \text{otherwise} \end{bmatrix}$$

At $t = 0^-$, the circuit is in the steady-state, so the inductor acts as a short-circuit and the capacitor acts as an open-circuit.

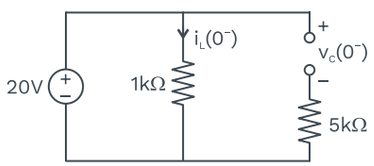


Fig. 6.29

$$i_L(0^-) = \frac{20}{1\times 10^3} = 20\text{mA}$$

$$v_C(0^-) = 20\text{V}$$

For t > 0 :

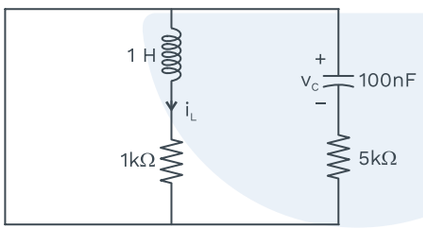


Fig. 6.30

Both L and C discharge independently through R in their branch and short-circuits.

Time constant:

$$\tau_1 = \frac{L}{R} = \frac{1}{1\times 10^3} = 1\,\text{m sec}$$

$$\tau_2 = RC = 5 \times 10^3 \times 100 \times 10^{-9}$$
$$= 500 \times 10^{-6}$$
$$= 0.5\ \text{m sec}$$

$$i_L = i(0)e^{-\frac{t}{\tau_1}}$$

$$= 20\,e^{-\frac{t}{1\times 10^{-3}}}\ \text{mA}$$

$$= 20e^{-1000t}\ \text{mA}$$

$$v_c = v_c(0)e^{-\frac{t}{\tau_2}}$$

$$= 20e^{-\frac{t}{0.5\times 10^{-3}}}$$

$$= 20e^{-2000t}\ \text{Volts}$$

Previous Years' Question

A DC voltage source is connected across a series RLC circuit. Under steady-state conditions, the applied DC voltage drops entirely across the

a) R only
b) L only
c) C only
d) R and L combination

Example 7:

For the circuit shown in the figure. Find the $\frac{di(0^+)}{dt}$.

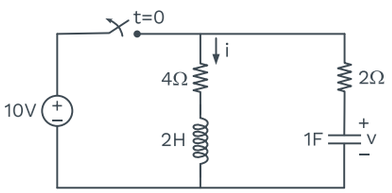


Fig. 6.31

Sol:

For t < 0, the switch is closed, and the circuit has reached a steady state.
The equivalent circuit is redrawn as shown below.

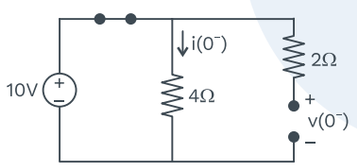


Fig. 6.32

$$i(0^-) = \frac{10}{4} = 2.5\ A$$

$V(0^-) = 10V$

At t = 0+;

$i(0^-) = i(0^+) = 2.5\ A$

$V(0^-) = V(0^+) = 10V$

For t > 0, the switch is open, and the circuit is redrawn as shown below.

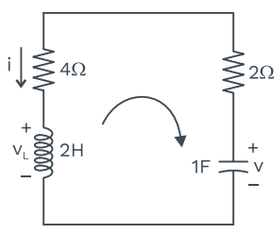


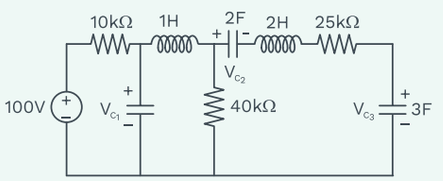
Fig. 6.33

$$V_L = L\frac{di}{dt}$$

$$\frac{di}{dt} = \frac{V_L}{L}$$

Previous Years' Question

The voltages V_{C_1}, V_{C_2} and V_{C_3} across the capacitors in the circuit in the figure, under steady-state, are respectively



a) 80V, 32V, 48V
b) 80V, 48V, 32V
c) 20V, 8V, 12V
d) 20V, 12V, 8V

Applying KVL at $t = 0^+$ we have,

$$-V_L(0^+) - 6i(0^+) + V(0^+) = 0$$

$$V_L(0^+) = V(0^+) - 6i(0^+)$$

$$= 10 - 6 \times 2.5$$

$$= 10 - 15 = -5V$$

At $t = 0^+$,

$$\frac{di(0^+)}{dt} = \frac{V_L(0^+)}{L}$$

$$\frac{di(0^+)}{dt} = \frac{-5}{2}$$

$$\frac{di(0^+)}{dt} = -2.5\ A/s$$

Example 8:

For the circuit shown below. Determine the voltage v(t) for t > 0.

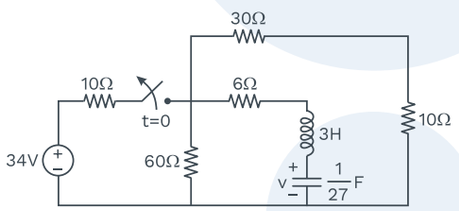


Fig. 6.34

Sol:

The 30Ω and 10 Ω resistors are in the series becomes 40 Ω, and combining the resistor 40 Ω and 60 Ω are in parallel is equal to 40 || 60 = 24 Ω.
Then, the circuit is equivalent to what is shown below.

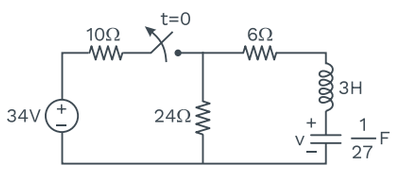


Fig. 6.35

At $t = 0^-$ (switch is closed), the circuit is in the steady-state.
So, the inductor becomes a short-circuit, and the capacitor becomes an open-circuit.
The circuit is redrawn, as shown below.

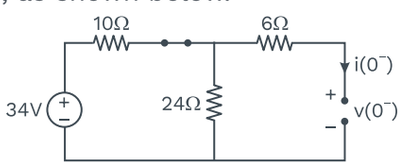


Fig. 6.36

$i(0^-) = 0A$

$i(0^-) = i(0) = i(0^+) = 0A$ ...(i)

$$v(0^-) = 34 \times \frac{24}{24+10} = 24V$$

$v(0^-) = v(0) = v(0^+) = 24V$ ...(ii)

For t > 0 (switch is open), we have a series RLC circuit as shown below.

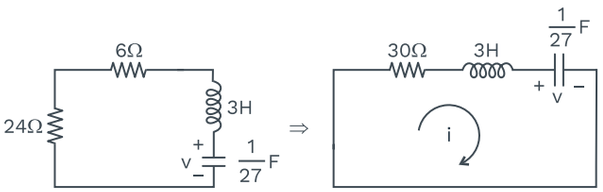


Fig. 6.37

$R = 30\Omega$, $L = 3H$, $C = \frac{1}{27}F$

For series RLC circuit $\Rightarrow s^2 + \frac{R}{L}s + \frac{1}{LC} = 0$

$$s_{1,2} = -\alpha \pm \sqrt{\alpha^2 - \omega_0^2}$$

where $\alpha = \frac{R}{2L}$, $\omega_0 = \frac{1}{\sqrt{LC}}$

$$\alpha = \frac{30}{2 \times 3} = 5, \quad \omega_0 = \frac{1}{\sqrt{3 \times \frac{1}{27}}} = 3 \text{ rad / sec}$$

$\alpha > \omega_0$ (overdamped response)

$$\therefore \quad s_{1,2} = -5 \pm \sqrt{5^2 - 3^2}$$

$$s_{1,2} = -5 \pm \sqrt{25 - 9} = -5 \pm 4$$

$$s_1 = -5 + 4 = -1$$

$$s_2 = -5 - 4 = -9$$

$$\therefore \quad v(t) = \left[Ae^{s_1 t} + Be^{s_2 t} \right]$$

$$\therefore \quad v(t) = \left[Ae^{-t} + Be^{-9t} \right]$$ ...(iii)

At t = 0,

$v(0) = Ae^0 + Be^0 = A + B$

$24 = A + B$

Or $A + B = 24$ ...(iv)

From the circuit, $i = C\frac{dv}{dt}$

$$i(t) = C\frac{d}{dt}\left[Ae^{-t} + Be^{-9t}\right]$$

$$i(t) = C[-Ae^{-t} - 9B\, e^{-9t}]$$

At t = 0,

$$i(0) = C\,[-A - 9B]$$

$$0 = -A - 9B$$

Or $A = -9B$ ...(v)

Substituting equation v) in the equation iv), we get:

$$-9B + B = 24$$

$$-8B = 24$$

$$B = -3$$

Putting the value of B in equation v), we get:

$$A = -9\times(-3) = 27$$

Putting the value of A and B in equation iii), we get,

$$v(t) = (27e^{-t} - 3e^{-9t})\ V$$

Example 9:

If the values of an inductor and a resistor are L = 1.5 H, R = 50Ω. What will be the value of C that will make an RLC series circuit overdamped?

Sol:

For series RLC circuit,

$$s^2 + \frac{R}{L}s + \frac{1}{LC} = 0$$

$$s = \frac{-\frac{R}{L} \pm \sqrt{\frac{R^2}{L^2} - \frac{4}{LC}}}{2}$$

$$= -\frac{R}{2L} \pm \sqrt{\frac{R^2}{4L^2} - \frac{1}{LC}}$$

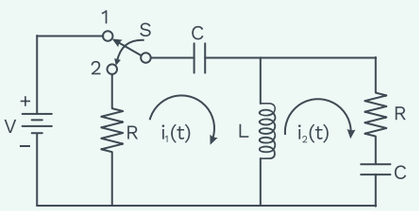
For overdamped, $\frac{R^2}{4L^2} > \frac{1}{LC}$

$$C > \frac{4L^2}{R^2L} \qquad C > \frac{4L}{R^2} = \frac{4\times1.5}{50\times50} = \frac{6}{2500} = 2.4\,mF$$

$\therefore$ C > 2.4 mF

Previous Years' Question

Assume that the switch S is in position-1 for a long time and thrown to position-2 at t = 0.



At $t = 0^+$, the current i_1 is:

a) $\frac{-V}{2R}A$ **b)** $\frac{-V}{R}A$

c) $\frac{-V}{4R}A$ **d)** Zero

Example 10:
Find v(t) for t > 0 in the circuit shown below.

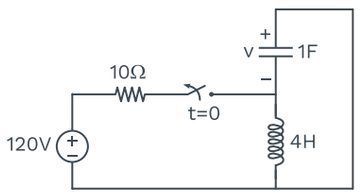


Fig. 6.38

Sol:
For t < 0, the equivalent circuit is redrawn as shown below.

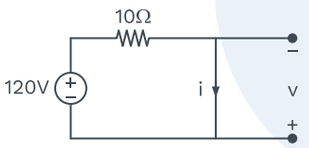


Fig. 6.39

$$i(0) = i(0^-) = i(0^+) = \frac{120}{10} = 12A,$$

$$v(0) = v(0^-) = v(0^+) = 0\,V$$

For t > 0, we have series RLC circuit as shown in the figure with R = 0 = α.

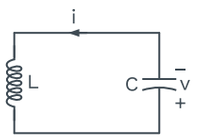


Fig. 6.40

$$\omega_0 = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{4\times 1}} = 0.5 = \omega_d$$

$$i(t) = [A\cos\omega_d t + B\sin\omega_d t]$$

$$i(t) = A\cos 0.5\,t + B\sin 0.5\,t]$$

At t = 0 sec,

$$i(0) = A\cos 0 + B\sin 0 \qquad \{\cos 0 = 1\}$$

$$12 = A + 0$$

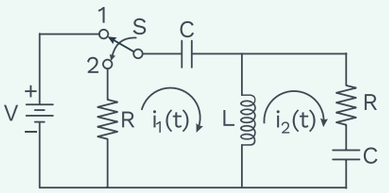
$$A = 12$$

$$\therefore\quad i(t) = 12\cos 0.5\,t + B\sin 0.5\,t$$

$$\frac{di(t)}{dt} = -6\sin 0.5t + 0.5B\cos 0.5t$$

Previous Years' Question

Assume that the switch S is in position-1 for a long time and thrown to position-2 at t = 0.



$I_1(s)$ and $I_2(s)$ are the Laplace transforms of $i_1(t)$ and $i_2(t)$ respectively. The equations for the loop currents $I_1(s)$ and $I_2(s)$ for the circuit shown in the figure, after the switch is brought from position 1 to position 2 at t = 0, are:

a) $\begin{bmatrix} R+Ls+\frac{1}{Cs} & -Ls \\ -Ls & R+\frac{1}{Cs} \end{bmatrix}\begin{bmatrix} I_1(s) \\ I_2(s) \end{bmatrix} = \begin{bmatrix} V/s \\ 0 \end{bmatrix}$

b) $\begin{bmatrix} R+Ls+\frac{1}{Cs} & -Ls \\ -Ls & R+\frac{1}{Cs} \end{bmatrix}\begin{bmatrix} I_1(s) \\ I_2(s) \end{bmatrix} = \begin{bmatrix} -V/s \\ 0 \end{bmatrix}$

c) $\begin{bmatrix} R+Ls+\frac{1}{Cs} & -Ls \\ -Ls & R+Ls+\frac{1}{Cs} \end{bmatrix}\begin{bmatrix} I_1(s) \\ I_2(s) \end{bmatrix} = \begin{bmatrix} -V/s \\ 0 \end{bmatrix}$

d) $\begin{bmatrix} R+Ls+\frac{1}{Cs} & -Ls \\ -Ls & R+Ls+\frac{1}{Cs} \end{bmatrix}\begin{bmatrix} I_1(s) \\ I_2(s) \end{bmatrix} = \begin{bmatrix} V/s \\ 0 \end{bmatrix}$

We know that, $v(t) = -L\frac{di(t)}{dt} \Rightarrow \frac{di(t)}{dt} = -\frac{v(t)}{L}$

$$-\frac{v(t)}{L} = -6\sin 0.5t + 0.5B\cos 0.5t$$

$v(t) = -4[-6 \sin 0.5\, t + 0.5\, B \cos 0.5t]$

$v(0) = 0 = -4[-6 \sin 0 + 0.5\, B \cos 0]$

$B = 0$

Thus, $v(t) = -4[-6 \sin 0.5\, t + 0]$

$v(t) = 24 \sin 0.5t\ V$

Example 11:

For the network shown below, what is the value of capacitor C needed, if the response is underdamped with 'unity damping factor' ($\alpha = 1$)?

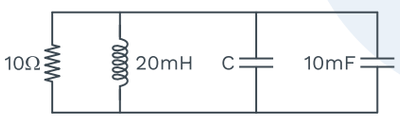


Fig. 6.41

Sol:

Let, $C_{eq} = C + 0.01$

For a parallel RLC circuit

$$\alpha = \frac{1}{2RC_{eq}},\ \omega_0 = \frac{1}{\sqrt{LC_{eq}}}$$

$$\alpha = 1 = \frac{1}{2 \times 10 \times C_{eq}}$$

Or, $C_{eq} = \frac{1}{20}$

Or, $C_{eq} = 0.05 = 50$ mF

$$\omega_0 = \frac{1}{\sqrt{0.02 \times 0.05}} = 31.62$$

$\omega_0 = 31.62 > \alpha$ (under damped)

$C_{eq} = C + 0.01$

$C = C_{eq} - 0.01$

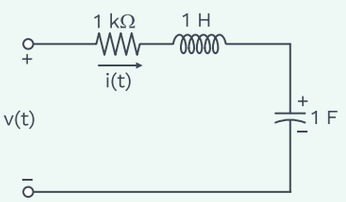
$= 0.05 - 0.01$

$= 0.04$ F

$= 40$ mF

Previous Years' Question

The circuit shown in the figure has an initial current $i_L(0^-)=1$ Amp through the inductor and an initial voltage $v_c(0^-)=-1$V across the capacitor. For input: v(t)=u(t), the Laplace transform of the current i(t) for $t \geq 0$ will be



a) $\frac{s}{s^2+s+1}$

b) $\frac{s+2}{s^2+s+1}$

c) $\frac{s-2}{s^2+s+1}$

d) $\frac{s+1}{s^2+s+1}$

Example 12:

For the circuit shown below, find the value of $i(0^+)$, $\frac{di(0^+)}{dt}, \frac{dv(0^+)}{dt}$.

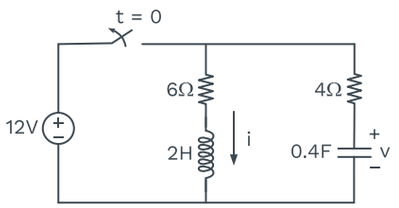


Fig. 6.42

Sol:

For t < 0, the circuit has reached steady state so that the equivalent circuit is shown in the figure.

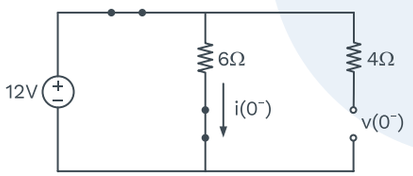


Fig. 6.43

$$i(0^-) = \frac{12}{6} = 2\,A, v(0^-) = 12\,V$$

$$i(0^-) = i(0) = i(0^+) = 2A$$

$$v(0^-) = v(0) = v(0^+) = 12\ V$$

For t > 0, we have the equivalent circuit in the figure.

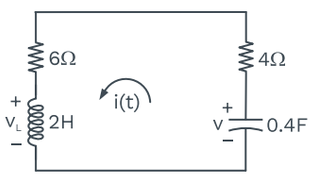


Fig. 6.44

Applying KVL at $t = 0^+$, we obtain,

$$-v_L(0^+) - 6i(0^+) - 4i(0^+) + v(0^+) = 0$$

$$v_L(0^+) = -10i(0^+) + v(0^+)$$

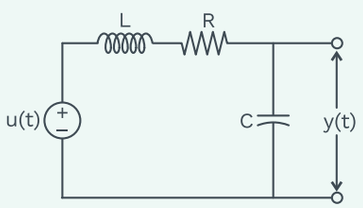
$$= -10 \times 2 + 12 = -20 + 12$$

$$v_L(0^+) = -8\ V$$

$$v_L(0^+) = L\frac{di(0^+)}{dt}$$

Previous Years' Question

The condition on R, L and C, such that the step response y(t) in the figure has no oscillations, is-



a) $R \geq \frac{1}{2}\sqrt{\frac{L}{C}}$ **b)** $R \geq \sqrt{\frac{L}{C}}$

c) $R \geq 2\sqrt{\frac{L}{C}}$ **d)** $R = \sqrt{\frac{1}{LC}}$

$$\frac{di(0^+)}{dt} = \frac{V_L(0^+)}{L} = \frac{-8}{2} = -4\ A/s$$

Similarly, $i_C = C\frac{dv}{dt}$ or $\frac{dv}{dt} = \frac{i_C}{C}$

$i_C(0^+) = -\ i(0^+) = -\ 2A$

At, $t = 0^+$

$$\frac{dv(0^+)}{dt} = \frac{i_C(0^+)}{C} = \frac{-2}{0.4} = -5\ V/s$$

At $t \to \infty$, the circuit reaches steady state.

$i(\infty) = 0A$, $v(\infty) = 0V$

Example 13:

For the circuit shown in the figure. Find the value of $i(\infty)$.

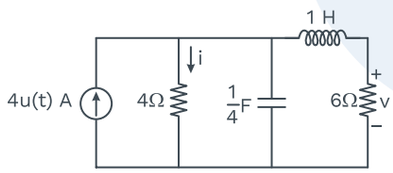


Fig. 6.45

Sol:

$$u(t) = \begin{cases} 0, & t < 0 \\ 1, & t > 0 \end{cases}$$

As $t \to \infty$, the circuit is in a steady-state.

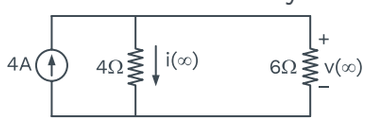


Fig. 6.46

$$i(\infty) = 4 \times \frac{6}{6+4} = 2.4\ A$$

Example 14:

For the electrical network shown below, determine the value of current i(t) for (t > 0).

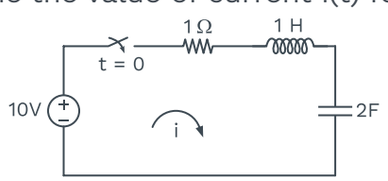
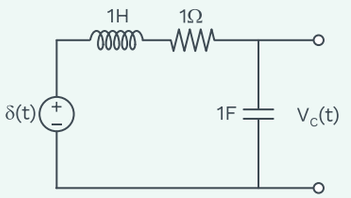


Fig. 6.47

Previous Years' Question

The following series RLC circuit with zero initial conditions is excited by a unit impulse function $\delta(t)$.



1) For t > 0, the output voltage $V_c(t)$ is:

a) $\frac{2}{\sqrt{3}}\left(e^{\frac{-1}{2}t} - e^{\frac{-\sqrt{3}}{2}t}\right)$

b) $\frac{2}{\sqrt{3}} te^{\frac{-1}{2}t}$

c) $\frac{2}{\sqrt{3}} e^{\frac{-1}{2}t} \cos\left(\frac{\sqrt{3}}{2}t\right)$

d) $\frac{2}{\sqrt{3}} e^{\frac{-1}{2}t} \sin\left(\frac{\sqrt{3}}{2}t\right)$

2) For t > 0, voltage across the resistor is:

a) $\frac{1}{\sqrt{3}}\left(e^{\frac{-\sqrt{3}}{2}t} - e^{\frac{-1}{2}t}\right)$

b) $e^{\frac{-1}{2}t}\left[\cos\left(\frac{\sqrt{3}}{2}t\right) - \frac{1}{\sqrt{3}}\sin\left(\frac{\sqrt{3}}{2}t\right)\right]$

c) $\frac{2}{\sqrt{3}} e^{\frac{-1}{2}t} \sin\left(\frac{\sqrt{3}}{2}t\right)$

d) $\frac{2}{\sqrt{3}} e^{\frac{-1}{2}t} \cos\left(\frac{\sqrt{3}}{2}t\right)$

Sol:

For, t < 0;

$i_L(0^-) = i_L(0^+) = 0A$

$v_C(0^-) = v_C(0^+) = 0V$

For, t > 0;

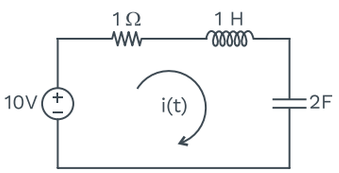


Fig. 6.48

Taking Laplace transform of the above circuit, we get,

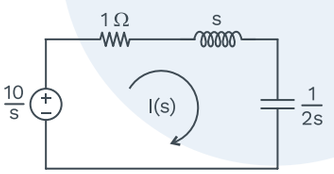


Fig. 6.49

$$I(s) = \frac{10/s}{1 + s + \frac{1}{2s}}$$

$$= \frac{10 \times 2}{2s + 2s^2 + 1} = \frac{20}{2s^2 + 2s + 1}$$

$$= \frac{10}{s^2 + s + 0.5}$$

$$= \frac{10}{s^2 + s + \frac{1}{4} + \frac{1}{4}}$$

$$= \frac{10}{\left(s + \frac{1}{2}\right)^2 + \left(\frac{1}{2}\right)^2} \times \left(\frac{\frac{1}{2}}{\frac{1}{2}}\right)$$

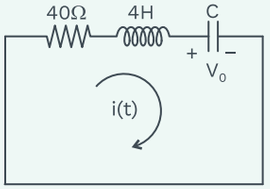
By taking the inverse Laplace transform, we get:

$$i(t) = 20\,e^{-t/2} \sin\frac{t}{2}$$

Or, $i(t) = 20e^{-0.5t} \sin 0.5t$ A

Previous Years' Question

In the circuit shown in the figure, the value of capacitor C (in mF) needed to have critically damped response i(t) is _____.



Example 15:
For the circuit shown below, find the $v_0(t)$ for t > 0.

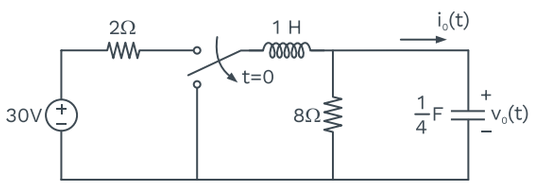


Fig. 6.50

Sol:
For t < 0,

$$i_L(0^-) = i_L(0^+) = \frac{30}{10} = 3A$$

$i_0(0^-) = 0A$

$$v_C(0^-) = v_C(0^+) = \frac{30 \times 8}{8+2} = 24\,V$$

For t > 0:

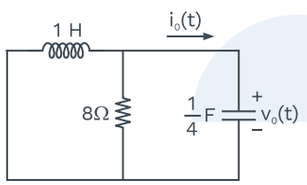


Fig. 6.51

By taking Laplace transform of the above circuit, we get:

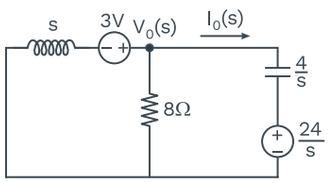


Fig. 6.52

[Voltage source in series with inductor L $i(0^+) = 1 \times 3 = 3V$]
Applying nodal analysis at node $V_0(s)$ in the above circuit, we get:

$$\frac{V_0(s)-3}{s} + \frac{V_0(s)}{8} + \frac{V_0(s)-24/s}{4/s} = 0$$

$$V_0(s)\left[\frac{1}{s} + \frac{1}{8} + \frac{s}{4}\right] = \frac{3}{s} + 6$$

$$V_0(s) = \frac{(6s+3)/s}{(8+s+2s^2)/8s} = \frac{8(6s+3)}{2s^2+s+8}$$

$$= \frac{24(2s+1)}{2s^2+s+8}$$

$$= 24\left[\frac{2s}{2s^2+s+8} + \frac{1}{2s^2+s+8}\right]$$

$$= 24\left[\frac{s}{s^2+0.5s+4} + \frac{0.5}{s^2+0.5s+4}\right]$$

$$= 24\left[\frac{s}{\left(s+\frac{1}{4}\right)^2+(1.9843)^2} + \frac{0.5}{\left(s+\frac{1}{4}\right)^2+(1.9843)^2}\right]$$

$$= 24\left[\frac{s+0.25-0.25}{\left(s+\frac{1}{4}\right)^2+(1.9843)^2} + \frac{0.5}{\left(s+\frac{1}{4}\right)^2+(1.9843)^2}\right]$$

$$= 24\left[\frac{s+0.25}{\left(s+\frac{1}{4}\right)^2+(1.9843)^2} + \frac{0.25}{\left(s+\frac{1}{4}\right)^2+(1.9843)^2}\right]$$

$$= 24\left[\frac{s+0.25}{\left(s+\frac{1}{4}\right)^2+(1.9843)^2} + \frac{0.25}{1.9843} \times \frac{1.9843}{\left(s+\frac{1}{4}\right)^2+(1.9843)^2}\right]$$

By taking the inverse Laplace transform, we get:

$$v_0(t) = 24\left[e^{-\frac{t}{4}}\cos 1.9843t + 0.1259e^{-\frac{t}{4}}\sin 1.9843t\right]$$

$$v_0(t) = \left[24\cos 1.9843t + 3.023\sin 1.9843t\right]e^{-\frac{t}{4}}\ V$$

Example 16:

For the circuit shown below, determine the value of resistance 'R' needed to make the response critically damped.

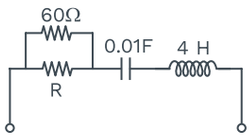


Fig. 6.53

Sol:

$$R_{TH} = R \parallel 60 = \frac{R \times 60}{R + 60}$$

For a series RLC circuit,

$$\omega_0 = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{4 \times 0.01}} = 5 \text{ rad/sec}$$

For critically damped,

$$\omega_0 = \alpha = \frac{R_{Th}}{2L} \qquad \left[\left(\frac{R_{TH}}{2L}\right)^2 = \frac{1}{LC}\right]$$

$$R_{TH} = 2L\,\omega_0$$

$$\frac{R \times 60}{R + 60} = 2 \times 4 \times 5$$

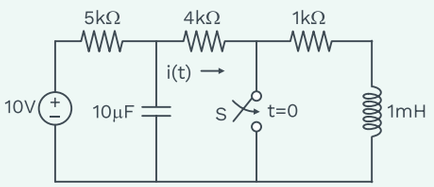
$$3R = 2(R + 60)$$

$$3R = 2R + 120$$

$$R = 120\ \Omega$$

Previous Years' Question

In the figure shown, the ideal switch has been open for a long time. If it is closed at t=0, then the magnitude of the current (in mA) through the 4kΩ resistor at $t = 0^+$ is _____.



Example 17:

For the circuit shown below, find the value of $V_L(0^+)$.

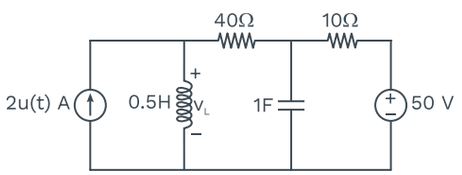


Fig. 6.54

Sol:

For t < 0;

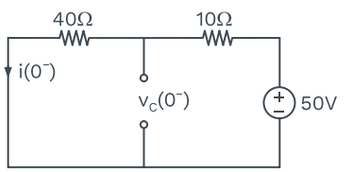


Fig. 6.55

$$i(0^-) = i(0^+) = \frac{50}{40+10} = 1\,A$$

$$v_C(0^-) = v_C(0^+) = \frac{50 \times 40}{40+10} = 40\,V$$

For $t = 0^+$;

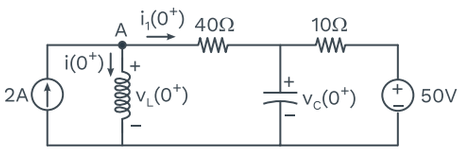


Fig. 6.56

Applying KCL at node A, we get,

$2 = i_1(0^+) + i(0^+) = i_1(0^+) + 1$

$i_1(0^+) = 1$ A

By KVL, we get-

$-v_L(0^+) + 40\, i_1(0^+) + v_C(0^+) = 0$

$v_L(0^+) = 40 \times 1 + 40 = 80$ V

Example 18:

What will be the value of R that will produce an undamped sinusoidal zero-input response in a parallel RLC circuit?

Sol:

For a parallel RLC circuit,

We know that,

$$\alpha = \frac{1}{2RC}$$

For undamped sinusoidal response i.e. $\alpha = 0$

$$\alpha = 0 = \frac{1}{2RC}$$

$R = \infty\ \Omega$

Example 19:

The voltage across the capacitor and current through the inductor in a series, RLC circuit is given below.

$$v_C(t) = 40 - 10e^{-20t} + 30e^{-10t}\ V$$

$$i_L(t) = 40e^{-20t} - 60e^{-10t}\ mA$$

Find the values of parameters R, L and C.

Sol:

Given,

$v_C(t) = 40 - 10e^{-20t} + 30e^{-10t}$ V

$i_L(t) = 40e^{-20t} - 60e^{-10t}$ mA

In a series RLC circuit,

$$i_L(t) = i_C(t) = C\frac{dv_C(t)}{dt}$$

$(40e^{-20t} - 60e^{-10t}) \times 10^{-3} = C[0 + 200e^{-20t} - 300e^{-10t}]$

Comparing equal power of exponential, we get,

$200\ C = 40 \times 10^{-3}$

$$C = \frac{40 \times 10^{-3}}{200}$$

$= 2 \times 10^{-4}$ F $= 200\ \mu$F

Let s_1 and s_2 are the roots of the characteristic equation.

$$s_1 = -\alpha + \sqrt{\alpha^2 - \omega_0^2} = -20$$

$$s_2 = -\alpha - \sqrt{\alpha^2 - \omega_0^2} = -10$$

$s_1 + s_2 = -2\alpha = -30$

Or $\alpha = 15$

$s_1 s_2 = \omega_0^2 = 200$

$$\omega_0^2 = \frac{1}{LC} = 200$$

$$L = \frac{1}{C \times 200}$$

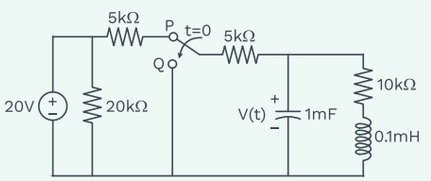
$$= \frac{1}{200 \times 10^{-6} \times 200} = 25\,H$$

$$\alpha = \frac{R}{2L} = 15$$

$R = 2 \times 15 \times 25 = 750\ \Omega$

Previous Years' Question

The switch in the circuit in the figure is in position P for a long time and then moved to the position Q at time t = 0.



The value of $\frac{d}{dt}V(t)$ at $t = 0^+$ is-

a) 0 V/sec **b)** 3 V/sec

c) −3 V/sec **d)** −5 V/sec

Chapter Summary



1) For finding the initial values x(0), $\frac{dx(0)}{dt}$ and a final value x(∞) for the analysis of second-order circuits are important.

2) If there are no independent sources present in the circuit after switching (or sudden change), then the circuit is a source-free circuit. The complete solution is the natural response.

3) The characteristic of second-order RLC circuit is $s^2 + 2\alpha s + \omega_0^2 = 0$. For a series circuit, $\alpha = R / 2L$, for a parallel circuit $\alpha = 1 / 2RC$, and for both cases $\omega_0 = 1 / \sqrt{LC}$, where α = Damping factor and ω_0 = Undamped natural frequency.

4) The natural response of an RLC circuit may be overdamped, under-damped, or critically damped, depending on the roots of the characteristic equation.

i) When the roots of the characteristic equation are equal $(\alpha = \omega_0$ or $s_1 = s_2)$, then the response is critically damped.

ii) When the roots of the characteristic equation are real and unequal $(s_1 \neq s_2$ or $\alpha > \omega_0)$, then the response is overdamped.

iii) When the roots are complex and unequal $(s_1 = s_2^*$ or $\alpha < \omega_0)$, then the response is underdamped.

5) The complete response is the sum of the transient and steady-state responses, if independent sources are present in the circuit after switching.

6) Series RLC (without source): $\alpha = \frac{R}{2L}$, $\omega_d = \sqrt{\omega_0^2 - \alpha^2}$

Underdamped ($\alpha < \omega_0$)

$$x(t) = e^{-\alpha t} (C_1 \cos \omega_d t + C_2 \sin \omega_d t)$$

- Critically damped ($\alpha = \omega_0$)

$$x(t) = e^{-\alpha t} (C_1 + C_2 t)$$

- Over damped ($\alpha > w_0$), roots = s_1, s_2

$$x(t) = C_1 e^{s_1 t} + C_2 e^{s_2 t}$$

7) Parallel RLC (without source): $\alpha = \frac{1}{2RC}$, $\omega_d = \sqrt{\omega_0^2 - \alpha^2}$

- Underdamped ($\alpha < \omega_0$)

$$x(t) = e^{-\alpha t} (C_1 \cos \omega_d t + C_2 \sin \omega_d t)$$

- Critically damped ($\alpha = \omega_0$)

$$x(t) = e^{-\alpha t}(C_1 + C_2 t)$$

- Over damped ($\alpha > \omega_0$), roots = s_1, s_2

$$x(t) = C_1 e^{s_1 t} + C_2 e^{s_2 t}$$

8) Series RLC (with source): $\alpha = \frac{R}{2L}$, $\omega_d = \sqrt{\omega_0^2 - \alpha^2}$

- Underdamped ($\alpha < \omega_0$)

$$x(t) = x(\infty) + e^{-\alpha t}(C_1 \cos \omega_d t + C_2 \sin \omega_d t)$$

- Critically damped ($\alpha = \omega_0$)

$$x(t) = x(\infty) + e^{-\alpha t}(C_1 + C_2 t)$$

- Over damped ($\alpha > \omega_0$), roots = s_1, s_2

$$x(t) = C_1 e^{s_1 t} + C_2 e^{s_2 t}$$

9) Parallel RLC (with source): $\alpha = \frac{1}{2RC}$, $\omega_d = \sqrt{\omega_0^2 - \alpha^2}$

- Underdamped ($\alpha < \omega_0$)

$$x(t) = x(\infty) + e^{-\alpha t}(C_1 \cos \omega_d t + C_2 \sin \omega_d t)$$

- Critically damped ($\alpha = \omega_0$)

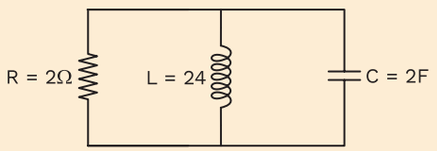
$$x(t) = x(\infty) + e^{-\alpha t}(C_1 + C_2 t)$$

- Over damped ($\alpha > \omega_0$), roots = s_1, s_2

$$x(t) = x(\infty) + C_1 e^{s_1 t} + C_2 e^{s_2 t}$$

PRACTICE QUESTIONS

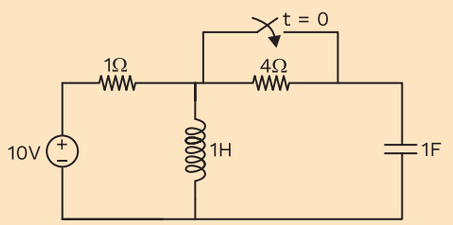
Q1 **The parallel RLC circuit is shown below, what type of response will it produce?**



a) Critically-damped **b) Under-damped**
c) Over-damped **d) None of the above**

Q2 **In an electric circuit, the dual conductance is:**
a) Resistance **b) Capacitance**
c) Short-circuit **d) Open-circuit**

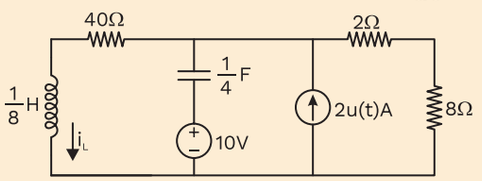
Q3 **For the circuit shown below, the initial inductor current (at t = 0) is:**



a) 1 A **b) 0 A** **c) 20 A** **d) 10 A**

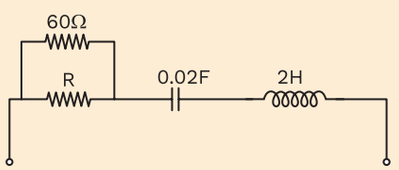
Q4 **A parallel RLC circuit has L = 1 H and C = 0.5 F. Determine the value of resistance 'R' that will produce an unity-damping factor is:**
a) 2 Ω **b) 5 Ω** **c) 1 Ω** **d) 6 Ω**

Q5 For the circuit shown below. Find the value of $\frac{di_L(0^+)}{dt}$.



a) 2 A/s b) 0 A/s c) 5 A/s d) 3 A/s

Q6 For the circuit shown below. Determine the value of resistance 'R' needed to make a critically-damped response.



a) 30 Ω b) 180 Ω c) 120 Ω d) 2 Ω

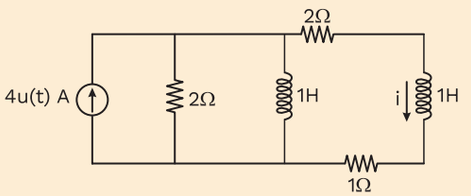
Q7 The step-response of an RLC circuit is given below:

$$\frac{d^2i}{dt} + 2\frac{di}{dt} + 5i = 5$$

Given that : i(0) = 1A and $\frac{di(0)}{dt} = 2$ A / s , determine the current i(t).

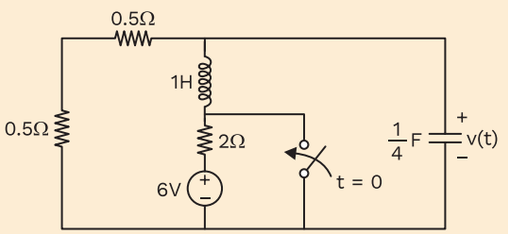
a) $2 + \sin 4t\, e^{-0.5t}$ A
b) $1 + 0.5 \sin 4t\, e^{-t}$ A
c) $1 + \sin 4t\, e^{-0.5t}$ A
d) $1 + 0.5 \sin 4t\, e^{-t}$ A

Q8 **For the circuit shown below. Determine the value of current 'i(t)' for t > 0.**



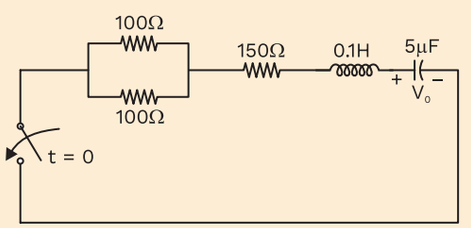
a) $3(e^{-t} + e^{-3t})$ A
b) $10e^{-t}$A
c) $2(e^{-t} + e^{-3t})$ A
d) $1.6(e^{-t} - e^{-6t})$A

Q9 **The circuit is shown below. Find voltage v(t).**



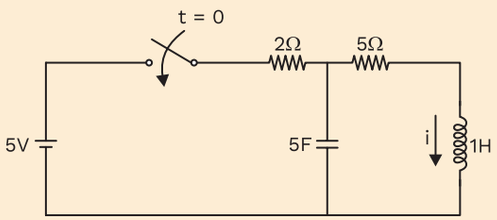
a) $4e^{-4t}$ V
b) $3 + 2e^{-t}$ V
c) $2(1 + 2t]\ e^{-2t}$ V
d) $-2 + 2e^{-t}$ V

Q10 **The series RLC circuit is shown below. Find the current transient after the switch is closed at t = 0. The initial energy stored in the capacitor is v_0 = 200 V.**



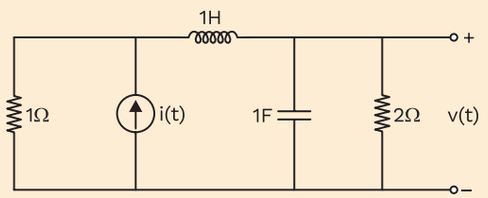
a) $5e^{-2t}$ sin 40t A
b) $1 + e^{-1000t}$ sin 1000t A
c) $\pm\ 2e^{-1000t}$ sin 1000 t A
d) None of these

Q11 For the network shown below, if the switch is closed at t = 0. Find $\frac{d^2i(0^+)}{dt^2}$.



a) 0.5 A/sec^2 b) 3 A/sec c) 2 A/sec^2 d) –2 A/sec^2

Q12 Find the differential equation of v(t) in the network shown below.



a) $\frac{d^2v(t)}{dt^2}+\frac{3}{2}\frac{dv(t)}{dt}+\frac{3}{2}v(t)=i(t)$

b) $\frac{d^2v(t)}{dt^2}-1.5\frac{dv(t)}{dt}+1.5v(t)=i(t)$

c) $\frac{d^2v(t)}{dt^2}+\frac{5}{6}\frac{dv(t)}{dt}+3v(t)=i(t)$

d) $\frac{d^2v(t)}{dt^2}+3\frac{dv(t)}{dt}+v(t)-i(t)=0$

Q13 **In the series RLC circuit, if the setting of resistance R = 0, it will produce.**
a) Underdamped response
b) Undamped response
c) Critically response
d) Overdamped response

Q14 **In the second-order RLC circuit, "critically damped response" peaks and decays:**
a) Faster than the overdamped response
b) Slower than the overdamped response
c) Same as an overdamped response
d) None of these

Q15 **The current in a series RLC network is given by:**

$$L\frac{d^2 i_L(t)}{dt^2} + R\frac{di_L(t)}{dt} + \frac{1}{C} i_L(t) = 0$$

If R = 8 W and C = 50 mF, then what value of L does make the response to be critically damped?
a) 0.8 H **b) 0.5 H** **c) 0.7 H** **d) 0.6 H**

Answer Key:

1)	b)	2)	a)	3)	d)	4)	c)	5)	b)
6)	c)	7)	b)	8)	d)	9)	c)	10)	c)
11)	a)	12)	a)	13)	b)	14)	a)	15)	a)